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Name:

**NATIONAL INSTITUTE OF TECHNOLOGY
KURUKSHETRA-136119
THEORY EXAMINATION**

Month and Year of the Exam: Aug 2026

Session: 2025-26

Programme: B.Tech

Semester: 4th

Course Code: EEPC211

Subject Title: DC&SM

Maximum Marks: 50

Time allowed: 3 Hours

Number of Questions to be attempted: 05

Unless stated otherwise, the symbols have their usual meanings in context with the subject.

Assume suitably and state, additional data required, if any. Graph paper may be provided.

The candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the Question Paper of correct course code and the correct subject title. Immediately point the discrepancy, if any, to the Supervisor on duty and sort it out.

Q.No.1(a) What are different types of Synchronous Machine and their application. Also give the advantage of inside-out construction.

(b) Derive the expression of pitch factor and distribution factor.

Q.No.2(a) Derive and draw the expression of Power of a salient pole synchronous machine. Also explain why salient pole machines give power even if excitation fails.

(b) A 3300 V, star-connected synchronous motor has synchronous impedance of $0.4 + j5\Omega$ per phase. For excitation EMF of 4000 V and motor input power of 1000 kW at rated voltage, Calculate the line current and power factor.

Q.No.3 (a) Draw and explain characteristics of various types DC motors with equations and also write the applications.

(b) Explain different types of speed control methods of DC motors. Also explain constant torque and constant power drive.

Q.No.4 (a) Derive the expression for active and reactive power flow through a Synchronous machine at sending and receiving end.

(b) Explain the power factor correction of a synchronous motor using field excitation at a particular load. Also draw V-curve and inverted V-curve.

Q.No.5(a) Explain commutation process in a DC machine with figures. Also explain resistance commutation, delayed commutation and voltage commutation.

(b) A 6-pole lap-wound DC generator has 600 conductors on its armature. The flux per pole is 0.02 Wb. Calculate (i) the speed at which the generator must be run to generate 300 V. (ii) What would be the speed if the generator were wave wound.

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Q.No.6(a) Explain the construction of the DC machine with a diagram. Also explain various losses in a DC machine.

(b) Explain armature reaction in a DC machine. What are the disadvantages of armature reaction and how these effects can be minimize?

Q.No.7(a) Derive the induced EMF and Torque equation of a DC machine.

(b) A 4-pole, 500 V DC shunt motor has 720 wave-connected conductors in the armature. The full-load armature current is 60A and the flux per pole is 0.03 Wb. The armature resistance is 0.2Ω and brush drop is 1V per brush. Calculate the full-load speed of the motor.

Q.No.8 A 6.6kV, 3 phase, 50Hz, star-connected alternator gave the following data for open circuit, short circuit and full-load power factor tests:

I_f in A	3.2	5.00	7.50	10.00	14.00
E_r in kV	3.10	4.90	6.60	7.50	8.24
I_{sc} in A	500	778	1170	—	—
Z.p.f.terminal voltage in kV	—	1.85	4.24	5.78	7.00

per phase armature resistance is 0.2Ω . Calculate the voltage regulation at full load current of 500A at 0.8 p.f. lagging by using following methods:

(i) e.m.f. method

(ii) m.m.f method

(End of Question Paper)